Sanchay Narendra Gawande

sanchay823@gmail.com | 857-395-5752 | Boston, MA | LinkedIn: https://www.linkedin.com/in/sanchay-gawande/ |
GitHub: https://github.com/SanchayGawande | Portfolio: https://sanchaygawande.com/

SUMMARY

Software Engineer with 2+ years of experience building scalable backend systems and AI infrastructure. Skilled in Python, JavaScript, microservices, REST, GraphQL APIs, and AWS. Built LLM orchestration and RAG workflows, delivered event-driven ingestion that reduced latency by 40%, and tuned database and caching layers to improve performance by 30%-50% while keeping query times under 200 ms.

TECHNICAL SKILLS

- **Programming:** Python, JavaScript, Java, SQL
- **Backend:** FastAPI, Flask, Node.js, GraphQL, REST
- **Frontend:** React, Redux, Next.js, D3.js, HTML5, CSS3
- **Databases:** PostgreSQL, MongoDB, Redis, Firebase
- **Cloud and DevOps:** AWS Lambda, API Gateway, SQS, ECS, S3, EC2, CloudFront, Docker, Kubernetes, GitHub Actions, CloudWatch, Terraform
- **Testing & Tools:** Pytest, Jest, Cypress, Agile/Scrum, Jira, SDLC, Git
- **AI/ML:** XGBoost, LightGBM, TensorFlow Serving, ONNX Runtime, OpenAI API, Embeddings, DistilBERT
- **Agents and Orchestration:** LangChain, LangGraph, Tool Execution, Function Calling

WORK EXPERIENCE

Founding Machine Learning Engineer, Suno Analytics — January 2025 to Present | Remote, USA

- Owned the backend and the core multi-model LLM orchestration layer using FastAPI, serving as the primary engineering owner for the ERP platform AI capabilities from design to production.
- Developed a stateful RAG system using Pinecone and metadata-driven routing, enabling long-running AI assistants to maintain context across complex financial analysis workflows.
- Implemented automated evaluation and regression checks using Pytest for RAG retrieval and tool calling, improving response quality by 25% and reducing incorrect tool triggers by 30% across common workflows.
- Automated an asynchronous ingestion pipeline using AWS Lambda and SQS to merge multi-tab Excel workbooks, reducing user-facing ingestion latency by 40% compared to the legacy synchronous process.
- Established a tool-calling infrastructure (MCP server) integrating Claude and GPT-4, enabling real-time actions such as macro generation.
- Optimized AWS infrastructure using ECS and API Gateway for cost and reliability, and set up monitoring for P99 latency across the orchestration layer.
- Hardened the ingestion and orchestration flows with timeouts, retries, and idempotency, reducing pipeline failures by 45% and improving end-to-end processing throughput by 2.0x during peak load.
- Introduced production observability with CloudWatch dashboards, alerts, and runbooks, cutting incident triage time by 35% and improving P99 stability by 20% across critical endpoints.

Software Engineer Fellowship, Headstarter — July 2024 to January 2025 | Remote, USA

- Delivered multiple full-stack and AI-driven applications end-to-end using Python, FastAPI, Node.js, React, PostgreSQL, and AWS, shipping production-quality features under Agile weekly release cycles managed in Jira.
- Integrated AI/ML features including LLM-powered components, inference workflows, and data processing services into application architectures to ensure stable and
- predictable execution.
- Led peer code reviews, provided PR-level feedback, and enforced coding practices that improved code quality and reduced debugging time across the cohort.
- Created internal utilities, debugging scripts, and deployment helpers that accelerated development velocity for multi-week team projects.
- Completed recurring engineering challenges with strict deadlines that replicated real production sprints, consistently ranking as a top performer in the fellowship.
- Shipped 8 production-style releases across 5 full-stack applications, delivering new features every week while maintaining consistent build and deploy hygiene.
- Set up CI/CD automation using GitHub Actions, reducing deployment time by 50% and cutting recurring integration issues by 30% through repeatable checks and
- standardized workflows.
- Reviewed 40+ pull requests and drove consistent code standards, reducing rework cycles by 25% and accelerating peer delivery during weekly sprints.

Software Engineer Intern, University of Massachusetts Boston — January 2024 to May 2024 | Boston, USA

- Architected a decoupled patient guidance platform, separating the React frontend from the Python inference backend to enable independent scaling and faster iteration.
- Implemented a hybrid search engine combining vector retrieval with metadata rules to structure nursing content, reducing LLM hallucinations in outputs.
- Collaborated in a 4-person cross-functional team to ship iConcern, a portal for diabetes-related patient education and support, aligning backend services with frontend and ML workflows.
- Owned the backend authentication track, implementing secure sign-in and session-handling patterns to protect user data and enable 3 core capabilities: educational articles, personalized chat assistant, and secure authentication.

PROJECTS

PeerGenius - AI-powered student collaboration platform

- Deployed a context-aware AI response system with 95% accuracy in academic query detection, differentiating solo vs group study modes, and using Firebase Authentication with JWT security.
- Developed a full-stack architecture with React 19, Express, MongoDB, and a CI/CD pipeline with production security standards.
- Implemented real-time participant and room management using Socket.IO and Mongoose ODM with role-based access control.

LifeLens - Microservices-based AI daily decision engine

- Built a full-stack microservices platform with Node.js, TypeScript, React Native, and FastAPI, integrating 5+ AI APIs with fallback mechanisms and 95%+ uptime.
- Optimized PostgreSQL (Row Level Security, indexing) and Redis caching, boosting performance by 30% to 50% via token optimization and cache invalidation while keeping query times under 200 ms.
- Launched a contextual AI nudge engine processing 7+ real-time factors (mood, weather, history) with a rule-based system to reduce user decision fatigue.

EDUCATION

University of Massachusetts Boston | Boston, MA, USA — Aug 2022 to May 2024
Master of Science in Computer Science

Sant Gadge Baba Amravati University | Maharashtra, India — June 2019 to May 2022
Bachelor of Engineering in Computer Science and Engineering